Examination/Reg No: **U15FF26S**

K. L. E SOCIETY'S

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B. C. A COLLEGE, ATHANI – 591304

B. C. A - I Semester

CERTIFICATE

This is to certify that Mr. / Miss. _____

Has completed the course of experiments in **C Programming LAB** prescribed by the Rani Channamma University, Belagavi for the **BCA-I** semester in the Laboratory of this College during the year 2026 – 2027.

Lab in Charge **Principal**

Examiners: 1. **2.**

C Programming LAB

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INDEX

Sl. No.	Program Name	Page No.	Sign
	PART A		
1	Write a program to read radius of a circle and to find and display area and circumference	1	
2	Write a program to read Principal amount, Time and Rate and calculate Simple Interest.	2	
3	Program to read three numbers and display the largest of the three.	3	
4	Write a program to function as a basic calculator; it should ask the user to input what type of arithmetic operation he would like, and then ask for the numbers on which the operation should be performed. The calculator should then give the output of the operation. Use switch. Error message should be reported, if any attempt is made to divide by zero.	4	
5	Write a program to find the roots of quadratic equation (Demonstration of else-if ladder)	5	
6	Write a program to read marks scored by a student in n subjects and find and display the average of all marks	6	
7	Write a program to demonstrate basic mathematical library functions defined in math.h	7	
8	Write a program to find HCF (GCD) of two numbers.	8	
9	Write a program that accepts a number 'n', and prints all prime numbers between 1 to n.	9	
10	Write a program to read a number, find the sum of its digits, reverse the number and check if it is a palindrome. Display appropriate message.	10	
11	Write a program to read numbers from keyboard continuously till the user presses 999 and to find the sum of only positive numbers. Display appropriate error messages.	11	
12	Write a program to read percentage of marks and to display appropriate message specifying class obtained. (demonstration of Switch Case statement)	12	
13	Write a program to read and concatenate two strings without using library function	13	
14	Write a program to print sum of even numbers and sum of odd numbers from array of integers which are to be inputed.	14	
15	Write a program to read a list of numbers, print it, then remove duplicate elements from the list and print the modified list. Use single dimensional Array	15	

Sl. No.	Program Name	Page No.	Sign
	PART B		
1	Write a program to swap(interchange) two numbers without using a temporary variable	16	
2	Write a program to read a string and to find the number of alphabets, digits, vowels, consonants, spaces and special characters using character functions	17-18	
3	Write a program to read a string, reverse it, concatenate the original string with its reverse, and print length of the original string and the concatenated string using built in functions	19	
4	Write a program to read a string and find the length of a string without using built in function	20	
5	Write a program to generate and display first n values in the Fibonacci sequence using recursive function	21	
6	Write a recursive function calculate factorial of a given integer, n.	22	
7	Write a program to read elements in a square matrix, display it in the form of a matrix and find and print its trace	23	
8	Write a program to read two matrices and perform addition and subtraction on them	24-25	
9	Write a program to read, display and multiply two m x n matrices using functions.	26-27	
10	Write a program to check and display if a number is prime by defining isprime() function	28	
11	Write a program using functions that takes in three arguments - a start temperature (in Celsius), an end temperature (in Celsius) and a step size. Print out a table that goes from the start temperature to the end temperature, in steps of the step size; converting each Celsius to Fahrenheit.	29	
12	Write a C program to create array of structure which stores Roll No, Name and Average marks of students. Accept n students data and print it in proper format.	30-31	
13	Write a C program to illustrate difference between structure and union by defining emp_no, emp_name, salary as members and display the size of the defined structure	32-33	

PART-A

1. Write a program to read radius of a circle and to find and display area and circumference

```
#include<stdio.h>
int main()
{
        float radius, area, circum;

        printf("Enter the radius of circle \n");

        scanf("%f",&radius);

        area=3.14*radius*radius;

        circum=2*3.14*radius;

        printf("area of circle: %f \n",area);

        printf("circumference of circle: %f \n",circum);

 }
```

Output:

```
Enter the radius of circle
10
area of circle: 314.000000
circumference of circle: 62.799999
```

2. Write a program to read Principal amount, Time and Rate and calculate Simple Interest.

```
#include <stdio.h>
int main()
{
   float p,r,t,SI;

   printf("Enter principal amount: ");
   scanf("%f", &p);

   printf("Enter interest rate: ");
   scanf("%f", &r);

   printf("Enter time period (in years): ");
   scanf("%f", &t);

   SI = (p * r * t) / 100;

   printf("Simple Interest is: %f\n", SI);

}
```

Output:

```
Enter principal amount: 1000
Enter interest rate: 5
Enter time period (in years): 1
Simple Interest is: 50.000000
```

3. Program to read three numbers and display the largest of the three.

```
#include <stdio.h>
int main( )
{
        int a,b,c;
        printf("Enter three numbers\n");
        scanf("%d%d%d",&a,&b,&c);

        if(a>b && a>c)
        {
                printf("%d is largest number\n",a);
        }
        else if(b>c)
        {
                printf("%d is largest number\n",b);
        }
        else
        {
                printf("%d is largest number\n",c);
        }
}
```

Output:

```
Enter values for a, b and c:
10 30 20
30 is the largest number.
```

4. Write a program to function as a basic calculator; it should ask the user to input what type of arithmetic operation he would like, and then ask for the numbers on which the operation should be performed. The calculator should then give the output of the operation. Use switch. Error message should be reported, if any attempt is made to divide by zero.

```
#include <stdio.h>
int main()
{
   char op;
   int first, second;
   printf("Enter an operator (+, -, *, /): ");
   scanf("%c", &op);
   printf("Enter two operands: ");
   scanf("%d %d", &first, &second);
   switch (op)
   {
      case '+':
      printf("%d + %d = %d", first, second, first + second);
      break;
      case '-':
      printf("%d - %d = %d", first, second, first - second);
      break;
      case '*':
      printf("%d * %d = %d", first, second, first * second);
      break;
      case '/':
      if(second == 0)
        {
                printf("Error! second number cannot be zero");
        }
      else
        {
                printf("%d / %d = %d", first, second, first / second);
        }
      break;
      // operator doesn't match any case constant
      default:
      printf("Error! operator is not correct");
   }
}
```

Output:

```
Enter an operator (+, -, *, /): +
Enter two operands: 12 15
12 + 15 = 27
```

5. Write a program to find the roots of quadratic equation (Demonstration of else-if ladder)

```
#include<stdio.h>
#include<math.h>
int main ()
{
   float a,b,c,r1,r2,d;
   printf ("Enter the values of a b c: ");
   scanf ("%f%f%f",&a,&b,&c);
   d= b*b - 4*a*c;
   if (d>0)
   {
      r1 = (-b+sqrt(d)) / (2*a);
      r2 = (-b-sqrt(d)) / (2*a);
      printf("The real roots = %.2f %.2f", r1, r2);
   }
   else if(d==0)
   {
      r1 = -b/(2*a);
      r2 = -b/(2*a);
      printf("Roots are equal =%f %f", r1, r2);
   }
   else
   {
      printf("Roots are imaginary");
   }
}
```

Output:

```
Enter the values of a b c: 1 -7 12
The real roots = 4.000000 3.000000
```

6. Write a program to read marks scored by a student in n subjects and find and display the average of all marks

```
#include<stdio.h>
int main()
{
   int i, n;
   float avg, mark, sum;

   printf("Enter number of subjects: ");
   scanf("%d",&n);

   printf("Enter Marks obtained in %d Subjects: \n",n);
   for(i=0; i<n; i++)
   {
      scanf("%f", &mark);
      sum = sum+mark;
   }

   avg = sum/n;

   printf("Average Marks = %f", avg);
}
```

Output:

```
Enter number of subjects: 3
Enter Marks obtained in 3 Subjects:
34
56
62
Average Marks = 50.666668
```

7. Write a program to demonstrate basic mathematical library functions defined in math.h

```
#include<stdio.h>
#include <math.h>

int main()
{
   printf("\n%f",ceil(3.6));
   printf("\n%f",ceil(3.3));
   printf("\n%f",floor(3.6));
   printf("\n%f",floor(3.2));
   printf("\n%f",sqrt(16));
   printf("\n%f",sqrt(7));
   printf("\n%f",pow(2,4));
   printf("\n%f",pow(3,3));
}
```

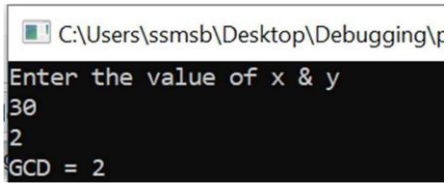
Output:

```
4.000000
4.000000
3.000000
3.000000
4.000000
2.645751
16.000000
27.000000
```

8. Write a program to find HCF (GCD) of two numbers.

```
#include<stdio.h>
int main()
{
	int x,y;
	printf("Enter the value of x & y\n");
	scanf("%d%d",&x,&y);
	while(x!=y)
	{
		if(x>y)
			x=x-y;
		else
			y=y-x;
	}
	printf("GCD = %d",x);
}
```

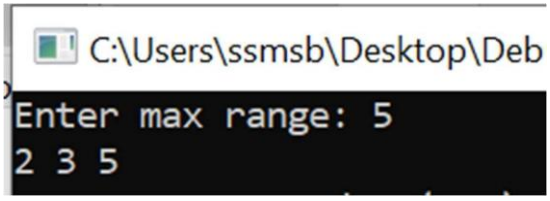
Output:



9. Write a program that accepts a number "n", and prints all prime numbers between 1 to n.

```
#include<stdio.h>
int main( )
{
	int i,j,count,n;
	printf("Enter max range: ");
	scanf("%d",&n);
	for(i = 1;i<=n;i++)
	{
		count = 0;
		for(j=2;j<=i/2;j++)
		{
			if(i%j==0)
			{
				count++;
				break;
			}
		}
		if(count==0 && i!= 1)
		printf("%d ",i);
	}
}
```

Output:



10. Write a program to read a number, find the sum of its digits, reverse the number and check if it is a palindrome. Display appropriate message.

```
#include <stdio.h>
int main()
{
   int n, temp, rem, rev = 0, sum=0;

   printf("Enter an number :");
   scanf("%d", &n);

   temp = n;
   while (n > 0)
   {
      rem = n % 10;
      rev = rev * 10 + rem;
      n = n / 10;
      sum = sum + rem;
    }
   printf("Given number is = %d\n", temp);
   printf("Sum of digits = %d\n", sum);
   printf("Its reverse is  = %d\n", rev);

   if (temp == rev)
   printf("Number is a palindrome \n");
   else
   printf("Number is not a palindrome \n");
}
```

Output:

```
Enter an integer :12321
Given number is = 12321
Sum of digits = 9
Its reverse is  = 12321
Number is a palindrome
```

11. Write a program to read numbers from keyboard continuously till the user presses 999 and to find the sum of only positive numbers. Display appropriate error messages.

```
#include<stdio.h>
int main()
{
   int n, sum=0;
   printf("Enter 999 to exit\n");

   while(1)
   {
      printf("Enter a number:");
      scanf("%d",&n);

      if(n==999)
      {
         printf("\nExiting the program");
         break;
      }
      else if(n>=0)
      {
         sum=sum+n;
      }
      else
      {
         printf("\nError: You entered negative Number");
      }
   }
   printf("\nSum of positive numbers = %d",sum);
}
```

Output:

```
Enter 999 to exit
Enter a number:23
Enter a number:21
Enter a number:45
Enter a number:999

Exiting the program
Sum of positive numbers = 89
```

12. Write a program to read percentage of marks and to display appropriate message specifying class obtained. (demonstration of Switch Case statement)

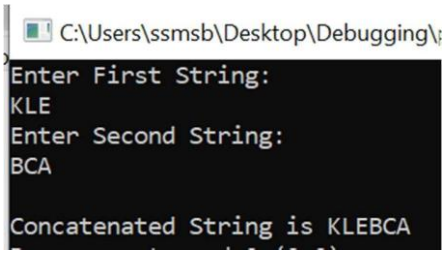
```
#include<stdio.h>
int main( )
{
 int marks;
 printf("Enter percentage of marks( 0-100 ): ");
 scanf("%d", &marks);
 switch( marks / 10 )
 {
   case 10:
   case 9:
         printf("Grade: A");
         break;
   case 8:
        printf("Grade: B");
         break;
   case 7:
         printf("Grade: C");
         break;
   case 6:
         printf("Grade: D");
         break;
   case 5:
         printf("Grade: E");
         break;
   case 4:
   case 3:
   case 2:
   case 1:
   case 0:
         printf("Grade: Fail");
         break;
   default:
         printf("Invalid Marks");
 }
}
```

```
Enter percentage of marks( 0-100 ): 90
Grade: A
```

13. Write a program to read and concatenate two strings without using library function

```
#include<stdio.h>
int main( )
 {
        char str1[25],str2[25];
        int i=0, j=0;
        printf("Enter First String: ");
        gets(str1);
        printf("Enter Second String:");
        gets(str2);
        while(str1[i]!='\0')
        i++;
        while(str2[j]!='\0')
        {
                str1[i]=str2[j];
                j++;
                i++;
        }
        str1[i]='\0';
        printf("\nConcatenated String is %s",str1);
}
```

Output:



14. Write a program to print sum of even numbers and sum of odd numbers from array of integers which are to be inputed.

```
#include<stdio.h>
int main()
{
   int a[10], n, i, sumeven=0, sumodd=0;

   printf("Enter array size: ");
   scanf("%d",&n);

   printf("Enter %d elements\n",n);
   for(i=0;i<n;i++)
   {
      scanf("%d",&a[i]);

      if(a[i]%2==0)
         sumeven = sumeven + a[i];
      else
         sumodd = sumodd + a[i];
   }
   printf("Sum of even numbers = %d",sumeven);
   printf("\nSum of odd numbers = %d",sumodd);
}
```

Output:

```
Enter array size: 5
Enter 5 elements
1 3 4 5 2
Sum of even numbers = 6
Sum of odd numbers = 9
```

15. Write a program to read a list of numbers, print it, then remove duplicate elements from the list and print the modified list. Use single dimensional Array

```
#include<stdio.h>
int main ( )
{
        int arr[20],i,j,k,n;
        printf("Enter the number of elements in an array:");
        scanf (" %d", &n);

        printf("Enter the elements of an array: \n");
        for ( i=0; i<n; i++)
        {
                scanf("%d",&arr[i]);
        }
        for(i=0;i< n;i++)
        {
                for(j=i+ 1;j<n;j++)
                {
                        if( arr[i]==arr[j])
                        {
                                for(k= j;k< n-1;k++)
                                {
                                        arr[k]=arr[k+ 1];
                                }
                                n--;
                                j--;
                        }
                }
        }
        printf("\n Array elements after deletion of the duplicate elements: ");
        for(i=0;i<n;i++)
        {
                printf("%d\t",arr[i]);
        }
}
```

Output:

```
Enter the number of elements in an array:5
Enter the elements of an array:
10
20
10
30
40
Array elements after deletion of the duplicate elements: 10 20 30 40
```

PART-B

1. Write a program to swap(interchange) two numbers without using a temporary variable

```
#include <stdio.h>
int main()
{
   int a, b;
   printf("Enter any two numbers: \n");
   scanf("%d%d", &a,&b);
   printf("Before swapping: a = %d and b = %d\n",a,b);
   a = a + b;
   b = a - b;
   a = a - b;
   printf("After swapping: a = %d and b = %d",a,b);
}
```

Output:

```
Enter any two numbers:
50
60
Before swapping: a = 50 and b = 60
After swapping: a = 60 and b = 50
```

2. Write a program to read a string and to find the number of alphabets, digits, vowels, consonants, spaces and special characters using character functions

```
#include  <stdio.h>
#include <ctype.h>
void main()
{
   char str[100];
   int i;
   int alphabets = 0, digits = 0, vowels = 0, consonants = 0, spaces = 0, specialChars = 0;
   printf("Enter a string: ");
   gets(str);
   for (i = 0; str[i] != '\0'; i++)
   {
      if (isalpha(str[i]))
      {
        alphabets++;
         if (str[i] == 'a' || str[i] == 'e' || str[i] == 'i' || str[i] == 'o' || str[i] == 'u' ||
            str[i] == 'A' || str[i] == 'E' || str[i] == 'I' || str[i] == 'O' || str[i] == 'U')
            {
                vowels++;
            }
             else
            {
                consonants++;
            }
      }
       else if (isdigit(str[i]))
      {
         digits++;
      } else if (isspace(str[i]))
      {
         spaces++;
      }
      else
      {
         specialChars++;
      }
   }
   printf("Alphabets: %d\n", alphabets);
   printf("Vowels: %d\n", vowels);
```

```
    printf("Consonants: %d\n", consonants);
    printf("Digits: %d\n", digits);
    printf("Spaces: %d\n", spaces);
    printf("Special Characters: %d\n", specialChars);
}
```

Output:

```
Enter a string: BCA College 123@
Alphabets: 10
Vowels: 4
Consonants: 6
Digits: 3
Spaces: 2
Special Characters: 1
```

3. Write a program to read a string, reverse it, concatenate the original string with its reverse, and print length of the original string and the concatenated string using built in functions

```
#include <stdio.h>
#include <string.h>
int main()
{
   char str[100], revstr[100], catstr[100];
   int lenstr, lencatstr;
   printf("Enter a string: ");
   gets(str);
        strcpy(revstr, str);
        strrev(revstr);
        strcpy(catstr, str);
        strcat(catstr, revstr);
        lenstr = strlen(str);
        lencatstr = strlen(catstr);
        printf("Original string: %s\n", str);
        printf("Reversed string: %s\n", revstr);
        printf("Concatenated string: %s\n", catstr);
        printf("Length of original string: %d\n", lenstr);
        printf("Length of concatenated string: %d\n", lencatstr);
}
```

OUTPUT:

```
Enter a string: BCA
Original string: BCA
Reversed string: ACB
Concatenated string: BCAACB
Length of original string: 3
Length of concatenated string: 6
```

4. Write a program to read a string and find the length of a string without using built in function

```
#include <stdio.h>
int main()
{
   char str[100];
   int length;
   printf("Enter a string: ");
   gets(str);
   length = 0;
   while (str[length] != '\0')
   {
      length++;
   }
   printf("Length of the string: %d\n", length);
}
```

Output:

```
Enter a string: BCA
Length of the string: 3
```

5. Write a program to generate and display first n values in the Fibonacci sequence using recursive function

```
#include <stdio.h>

int fibonacci(int n)
{
   if (n == 1)
   {
      return 0;
   }
   else if (n == 2)
   {
       return 1;
   }
         else
         {
                  return fibonacci(n - 1) + fibonacci(n - 2);
         }
}

int main()
{
   int n, i;
   printf("Enter the number of Fibonacci values to generate: ");
   scanf("%d", &n);
   printf("First %d values in the Fibonacci sequence:\n", n);
   for (i = 1; i <= n; i++)
   {
      printf("%d ", fibonacci(i));
   }
   printf("\n");
}
```

Output:

```
Enter the number of Fibonacci values to generate: 5
First 5 values in the Fibonacci sequence:
0 1 1 2 3
```

6. Write a recursive function calculate factorial of a given integer, n.

```
#include <stdio.h>

int factorial(int n)
{
   if (n == 0)
   {
      return 1;
   }
   else
   {
      return n * factorial(n - 1);
   }
}

int main()
{
   int number;
   printf("Enter an integer: ");
   scanf("%d", &number);
   printf("Factorial of %d is %d\n", number, factorial(number));
}
```

Output:

```
Enter an integer: 5
Factorial of 5 is 120
```

7. Write a program to read elements in a square matrix, display it in the form of a matrix and find and print its trace

```
#include <stdio.h>
int main()
{
   int n, i, j;
   printf("Enter the size of the square matrix (n): ");
   scanf("%d", &n);
   int matrix[n][n];
   printf("Enter the elements of the matrix:\n");
   for (i = 0; i < n; i++)
   {
      for (j = 0; j < n; j++)
      {
         scanf("%d", &matrix[i][j]);
      }
   }
   printf("The matrix is:\n");
   for (i = 0; i < n; i++) {
      for (j = 0; j < n; j++) {
         printf("%d ", matrix[i][j]);
      }
      printf("\n");
   }
   int trace = 0;
   for (i = 0; i < n; i++) {
      trace += matrix[i][i];
   }
   printf("The trace of the matrix is: %d\n", trace);
}
```

```
Enter the size of the square matrix (n): 2
Enter the elements of the matrix:
1 3
2 5
The matrix is:
1 3
2 5
The trace of the matrix is: 6
```

8. Write a program to read two matrices and perform addition and subtraction on them

```
#include <stdio.h>
void main()
{
   int rows, cols, i, j;
   int matrix1[10][10], matrix2[10][10];
   int sum[10][10], diff[10][10];
   printf("Enter the number of rows: ");
   scanf("%d", &rows);
   printf("Enter the number of columns: ");
   scanf("%d", &cols);
   printf("Enter elements of the first matrix:\n");
   for (i = 0; i < rows; i++)
   {
      for (j = 0; j < cols; j++)
      {
         scanf("%d", &matrix1[i][j]);
      }
   }
   printf("Enter elements of the second matrix:\n");
   for (i = 0; i < rows; i++)
   {
      for (j = 0; j < cols; j++)
      {
         scanf("%d", &matrix2[i][j]);
      }
   }
   for (i = 0; i < rows; i++)
   {
      for (j = 0; j < cols; j++)
      {
         sum[i][j] = matrix1[i][j] + matrix2[i][j];
         diff[i][j] = matrix1[i][j] - matrix2[i][j];
      }
   }
```

```
    printf("Sum of the matrices:\n");
    for (i = 0; i < rows; i++)
    {
        for (j = 0; j < cols; j++)
        {
            printf("%d ", sum[i][j]);
        }
        printf("\n");
    }
    printf("Difference of the matrices:\n");
    for (i = 0; i < rows; i++)
    {
        for (j = 0; j < cols; j++)
        {
            printf("%d ", diff[i][j]);
        }
        printf("\n");
    }
}
```

Output:

```
Enter the number of rows: 2
Enter the number of columns: 2
Enter elements of the first matrix:
1 3
4 5
Enter elements of the second matrix:
2 3
2 1
Sum of the matrices:
3 6
6 6
Difference of the matrices:
-1 0
2 4
```

9. Write a program to read, display and multiply two m x n matrices using functions.

```
#include <stdio.h>
void readMatrix(int matrix[][10], int rows, int cols);
void displayMatrix(int matrix[][10], int rows, int cols);
void multiplyMatrices(int matrix1[][10], int matrix2[][10], int result[][10], int
rows1, int cols1, int rows2, int cols2);
void main()
{
   int rows1, cols1, rows2, cols2;
   printf("Enter the number of rows and columns of the first matrix: ");
   scanf("%d %d", &rows1, &cols1);
   printf("Enter the number of rows and columns of the second matrix: ");
   scanf("%d %d", &rows2, &cols2);
   if (cols1 != rows2)
   {
      printf("Matrix multiplication is not possible.\n");
      return 1;
   }
   int matrix1[10][10], matrix2[10][10], result[10][10] = {0};
   printf("Enter elements of the first matrix:\n");
   readMatrix(matrix1, rows1, cols1);
   printf("Enter elements of the second matrix:\n");
   readMatrix(matrix2, rows2, cols2);
   multiplyMatrices(matrix1, matrix2, result, rows1, cols1, rows2, cols2);
   printf("First matrix:\n");
   displayMatrix(matrix1, rows1, cols1);
   printf("Second matrix:\n");
   displayMatrix(matrix2, rows2, cols2);
   printf("Resultant matrix after multiplication:\n");
   displayMatrix(result, rows1, cols2);
}
void readMatrix(int matrix[][10], int rows, int cols)
{
for (int i = 0; i < rows; i++)
{
      for (int j = 0; j < cols; j++)
      {
         scanf("%d", &matrix[i][j]);
      }
```

```
    }
}

void displayMatrix(int matrix[][10], int rows, int cols)
{
    for (int i = 0; i < rows; i++)
    {
        for (int j = 0; j < cols; j++)
        {
            printf("%d ", matrix[i][j]);
        }
        printf("\n");
    }
}
void multiplyMatrices(int matrix1[][10], int matrix2[][10], int result[][10], int
rows1, int cols1, int rows2, int cols2) {
    for (int i = 0; i < rows1; i++) {
        for (int j = 0; j < cols2; j++) {
            for (int k = 0; k < cols1; k++) {
                result[i][j] += matrix1[i][k] * matrix2[k][j];
            }
        }
    }
```

Output:

```
Enter elements of the first matrix:
1 3
4 2
Enter elements of the second matrix:
3 4
2 1
First matrix:
1 3
4 2
Second matrix:
3 4
2 1
Resultant matrix after multiplication:
9 7
16 18
```

10. Write a program to check and display if a number is prime by defining isprime( ) function

```
#include <stdio.h>
int isPrime(int num)
{
   if (num <= 1)
   {
      return 0;
   }
   for (int i = 2; i <= num / 2; i++)
   {
      if (num % i == 0)
      {
         return 0;
      }
   }
   return 1;
}
int main()
{
   int number;
   printf("Enter an integer: ");
   scanf("%d", &number);
   if (isPrime(number))
   {
      printf("%d is a prime number.\n", number);
   }
    else
   {
      printf("%d is not a prime number.\n", number);
   }
}
```

Output:

```
Enter an integer: 23
23 is a prime number.
```

11. Write a program using functions that takes in three arguments - a start temperature (in Celsius), an end temperature (in Celsius) and a step size. Print out a table that goes from the start temperature to the end temperature, in steps of the step size; converting each Celsius to Fahrenheit.

```
#include <stdio.h>
void printTemperatureTable(int start, int end, int step);
float celsiusToFahrenheit(float celsius);
void main()
{
   int startTemp, endTemp, stepSize;
   printf("Enter the start temperature (in Celsius): ");
   scanf("%d", &startTemp);
   printf("Enter the end temperature (in Celsius): ");
   scanf("%d", &endTemp);
   printf("Enter the step size: ");
   scanf("%d", &stepSize);
   printTemperatureTable(startTemp, endTemp, stepSize);
}
void printTemperatureTable(int start, int end, int step)
{ printf("Celsius\tFahrenheit\n");
   for (int celsius = start; celsius <= end; celsius += step)
   { printf("%d\t%.2f\n", celsius, celsiusToFahrenheit(celsius));
   }
}
float celsiusToFahrenheit(float celsius) {
   return (celsius * 9.0 / 5.0) + 32;
}
```

Output:

```
Enter the start temperature (in Celsius): 25
Enter the end temperature (in Celsius): 35
Enter the step size: 2
Celsius Fahrenheit
25      77.00
27      80.60
29      84.20
31      87.80
33      91.40
35      95.00
```

12. Write a C program to create array of structure which stores Roll No, Name and Average marks of students. Accept n students data and print it in proper format.

```
#include <stdio.h>
// Define a structure to store student details
struct Student
{
   int rollNo;
   char name[50];
   float avgMarks;
};

int main()
{
   // Declare an array of structures to store student data
   struct Student students[100];
   int n;
   // Read the number of students
   printf("Enter the number of students: ");
   scanf("%d", &n);
   // Read data for each student
   for (int i = 0; i < n; i++) {
      printf("Enter details for student %d:\n", i + 1);
      printf("Roll No: ");
      scanf("%d", &students[i].rollNo);
      printf("Name: ");
      scanf("%s", &students[i].name);
      printf("Average Marks: "); scanf("%f",
      &students[i].avgMarks);
   }
   // Display student data
   printf("\nStudent Details:\n");
   printf("Roll No\tName\t\tAverage Marks\n");
```

```
    printf("---------------------------------------- \n");
    for (int i = 0; i < n; i++) {
        printf("%d\t%s\t\t%.2f\n", students[i].rollNo, students[i].name,
students[i].avgMarks);
    }
}
```

Output:

```
Enter the number of students: 3
Enter details for student 1:
Roll No: 1001
Name: Anita
Average Marks: 76.5
Enter details for student 2:
Roll No: 1002
Name: Samay
Average Marks: 67.5
Enter details for student 3:
Roll No: 1003
Name: Vivek
Average Marks: 71.1

Student Details:
Roll No Name            Average Marks
----------------------------------------
1001    Anita           76.50
1002    Samay           67.50
1003    Vivek           71.10
```

13. Write a C program to illustrate difference between structure and union by defining emp_no , emp_name, salary as members and display the size of the defined structure

```
#include <stdio.h>
#include <string.h>

// Define a structure to store employee details
struct EmployeeStruct
{
   int emp_no;
   char emp_name[50];
   float salary;
};

// Define a union to store employee details
union EmployeeUnion
{
   int emp_no;
   char emp_name[50];
   float salary;
};

void main()
{
   // Declare variables of structure and union
   struct EmployeeStruct empStruct;
   union EmployeeUnion empUnion;

   // Assign values to structure members
   empStruct.emp_no = 1;
   strcpy(empStruct.emp_name, "John Doe");
   empStruct.salary = 50000.0;

   // Assign values to union members
   empUnion.emp_no = 2;
   strcpy(empUnion.emp_name, "Jane Doe");
   empUnion.salary = 60000.0;

   // Display the sizes of structure and union
   printf("Size of Employee structure: %lu bytes\n", sizeof(empStruct));
```

```
    printf("Size of Employee union: %lu bytes\n", sizeof(empUnion));
    // Display the members of structure
    printf("\nStructure Employee Details:\n");
    printf("Employee Number: %d\n", empStruct.emp_no);
    printf("Employee Name: %s\n", empStruct.emp_name);
    printf("Employee Salary: %.2f\n", empStruct.salary);

    // Display the members of union
    printf("\nUnion Employee Details:\n");
    printf("Employee Number: %d\n", empUnion.emp_no);
    printf("Employee Name: %s\n", empUnion.emp_name);
    printf("Employee Salary: %.2f\n", empUnion.salary);
}
```

Output:

```
Size of Employee structure: 60 bytes
Size of Employee union: 52 bytes

Structure Employee Details:
Employee Number: 1
Employee Name: John Doe
Employee Salary: 50000.00

Union Employee Details:
Employee Number: 1198153728
Employee Name:
Employee Salary: 60000.00
```